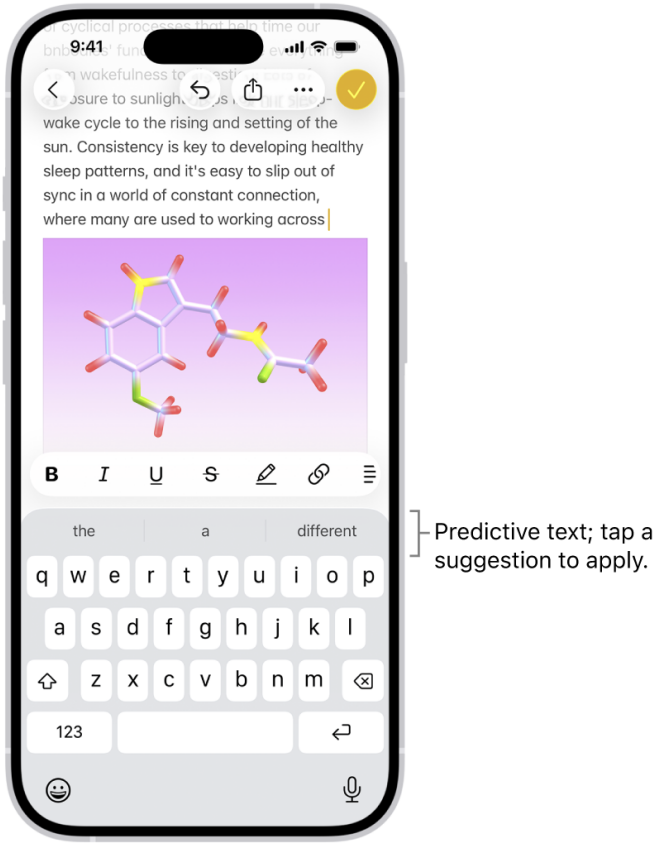
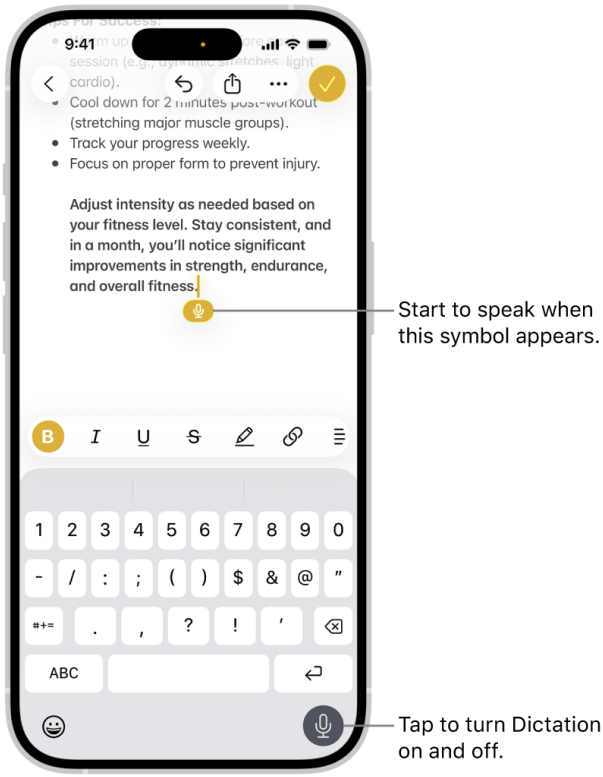
Predictive Text, Dictation, and Everyday Language AI

Portfolio Artifact Brief - Rashed Rabata

This artifact uses predictive text and dictation on iPhone as everyday examples of machine learning and deep learning. The screenshots provide tangible evidence of language-focused AI features that I use regularly, while the write-up connects those features to neural language models, automatic speech recognition, and practical communication workflows.

Title Predictive Text, Dictation, and Everyday Language AI	**Objective** Show how predictive text and dictation demonstrate applied machine learning through next-word prediction and speech-to-text on iPhone.
Tools iPhone screenshots, Predictive Text, Dictation, Apple Support documentation, Apple Machine Learning Research, GitHub Pages, HTML/CSS, PDF packaging.	**Value Proposition** Shows that I can connect familiar language tools to foundational AI concepts and explain them clearly with direct visual evidence.





Predictive text: Suggested words appear above the keyboard and can be tapped to apply the next word or phrase.

Dictation: The on-screen prompt and microphone button show how voice input is converted into text from within the keyboard workflow.

Predictive text is relevant to machine learning because it estimates likely next words from context rather than relying on a fixed static rule set. Dictation is relevant because speech recognition uses trained models to convert spoken audio into written text. Apple Support describes the user-facing features, while Apple Machine Learning Research provides the technical context through neural language models for keyboard prediction and neural automatic speech recognition for speech-to-text.

Process

1. Selected predictive text and dictation as everyday AI examples.
2. Captured annotated screenshots.
3. Reviewed Apple Support documentation and Apple machine learning research.
4. Connected the evidence to course concepts such as language modeling, neural networks, and speech recognition.
5. Packaged the material into a portfolio-ready artifact.

Relevance

This artifact is directly tied to coursework because it applies foundational concepts from machine learning and deep learning to a real product interface. It demonstrates meaningful use of the artifact template with clear evidence, a professional presentation, and a connection between daily technology use and core AI ideas.

APA 7 References

Apple. (2018, September 27). *Can global semantic context improve neural language models?* Apple Machine Learning Research. https://machinelearning.apple.com/research/can-global-semantic-context-improve-neural-language-models

Apple. (2024). *Enhancing CTC-based speech recognition with diverse modeling units*. Apple Machine Learning Research. https://machinelearning.apple.com/research/enhancing-ctc-speech

Apple Support. (n.d.). *Use predictive text on iPhone*. https://support.apple.com/guide/iphone/use-predictive-text-iphd4ea90231/ios

Apple Support. (n.d.). *Dictate text on iPhone*. https://support.apple.com/guide/iphone/dictate-text-iph2c0651d2/ios

Apple Support. (2025, February 4). *How to use auto-correction and predictive text on your iPhone, iPad, or iPod touch*. https://support.apple.com/en-us/104995